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ESCRITA CIENTÍFICA (ESTRUTURA)

Eduardo Ogasawara
eogasawara@ieee.org
https://eic.cefet-rj.br/~eogasawara

Estrutura geral de textos científicos em Computação

- Introdução
- Referencial teórico / Trabalhos relacionados
- Método
- Resultados
- Conclusões

[1] R. Wazlawick, 2017, *Metodologia de Pesquisa para Ciência da Computação*. Elsevier Brasil.
[2] J. Zobel, 2015, *Writing for Computer Science*. Springer.

A introdução

- Motivação (M)
  - Desafios em abertos com citações a trabalhos no tema
  - Costuma ter de três a seis parágrafos
- Definição do Problema (DP)
  - Objetivo central do trabalho
  - Pode ter objetivos específicos
    - Derivados do “detalhamento” do objetivo central
  - Costuma ter de um a três parágrafos
- Contribuição (C)
- Principais resultados (R)
- Estrutura (E)

Evolving Systems
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ORIGINAL PAPER

Analyzing flight delay prediction under concept drift

Lucas Giusti[1] · Leonardo Carvalho[1] · Antonio Tadeu Gomes[2] · Rafaelli Coutinho[1] · Jorge Soares[1] · Eduardo Ogasawara[1]

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Abstract
Flight delays impose challenges that impact any flight transportation system-predicting when they will occur in a meaningful way to mitigate this issue. However, the distribution of the flight delay system variables changes over time. This phenomenon is known in predictive analytics as concept drift. This paper investigates the prediction performance of different drift handling strategies in aviation under different scales (models trained from flights related to a single airport or the entire flight system). Specifically, two research questions were proposed and answered: (1) how do drift handling strategies influence the prediction performance of delays? (2) Do different scales change the results of drift handling strategies? In our analysis, drift handling strategies are relevant, and their impacts vary according to scale and machine learning models.

Keywords Flight delays · Prediction · Classification · Concept drift

1 Introduction

(M) Delay is one of the most critical indicators for flight transportation systems. Flight delays impose a challenge that impacts any flight transportation system. In the United States (US), it is estimated that a 10% decrease in flight delays would mean a US$ 8 billion (year base 2013) increase in gross domestic product (GDP) (Peterson et al. 2013). In this context, the prediction of delayed flights may be an essential tool for effectively addressing this problem.

(M) The development of machine learning models makes it possible to identify potentially delayed flights or critical periods before happening, enabling better planning. For that reason, many predictive models have been developed to achieve the task (Rong et al. 2015; Kim et al. 2016; Yu et al. 2019). Commonly, flight data is combined with weather information from departure and arrival locations to help predict flight delays (Du et al. 2018; Wu and Law 2019).

From the machine learning point of view, predicting delay may be a regression or classification task. In the former, the goal is to predict the time (usually measured in minutes) that a flight will delay. In the latter, the goal is to predict whether the flight is going to delay (Kim et al. 2016; Gui et al. 2020). The literature specialized in flight delay prediction provides many different models that have been developed with good results for both tasks. Specifically, for classification, which is the focus of the present study, random forest and deep recurrent neural network models have shown promising results in the US and China datasets (Rong et al. 2015; Yu et al. 2019).

(M) Large flight systems (such as the US, China, Europe, and Brazil) have challenges that impact flight delays. The relation of delays with input variables, such as destination or weather, may vary according to time and space (Sternberg et al. 2016). Thus, the proportion of delays may vary from time to time. Such a variation may occur due to punctual events such as storms and strikes. Other variations are disruptive, such as the FIFA World Cup in 2014. It led to an increase in the airports' capacity throughout most of the Brazilian Flight System and a significant change in flights and passengers.

(M) Nevertheless, even when there is no perceived change in the size of the flight system, the distributions of each feature and the relationship between system features and delays may vary. These relationship variations lead to a scenario called *concept drift* (Iwashita and Papa 2019). When the variables change but do not interfere with how delay occurs, there is no concept drift. Generally, a concept drift is a (statistically

✉ Eduardo Ogasawara
eogasawara@ieee.org

[1] Federal Center for Technological Education of Rio de Janeiro, CEFET/RJ, Rio de Janeiro, Brazil

[2] National Laboratory of Scientific Computing, LNCC, Petrópolis, Brazil

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(M) significant) difference between the joint probability of input and output variables observed in different dataset samples.

Previous studies indicate that concept drift may impact predictive models (Gama et al. 2014; Webb et al. 2016; Iwashita and Papa 2019). Considering the nature of evolving systems, some approaches applied soft computing methods to adapt to handle concept drifts (Angelov and Zhou 2008; Lughofer and Angelov 2011; Zhou and Angelov 2007). Other approaches tested algorithms that retrain the aviation models if drift is detected, like recurrent neural networks (Khamassi and Sayed-Mouchaweh 2014; Pesaranghader et al. 2016; Kim et al. 2016). Moreover, the amount of data used to train each model (so-called *scale*) varies across past studies: models trained from flights related to a single airport or the entire flight system. To the best of our knowledge, no study considered different system scales.

(DP) This paper investigates the prediction performance of different drift handling strategies in aviation under different scales. It considers two different scales: *system-based* (*SB*) and *airport-based* (*AB*). In (*SB*), all airports in the flight system are considered together. Conversely, in *AB*, each airport is taken into account separately. Specifically, two research questions were proposed and answered: (1) how do drift handling strategies influence the prediction performance of delays? (2) Do different scales change the results of drift handling strategies? These questions are answered while studying the Brazilian flight system dataset. It is an integrated database containing flight operations data provided by the Brazilian National Civil Aviation Agency (ANAC) (ANAC 2017) and airport weather data provided by automated surface observing systems (ASOS) (ASOS 2019). The main contributions to the state-of-the-art of the present work were the following:

(C/R)

- Always retraining the models may offer better recall with similar precision scores.
- Training one model for each airport show generally better predictive performance;
- Training with higher periods (1, 2, and 3 years) showed good results, but windows longer than 1 year are not significantly better for prediction performance;
- Random Forests and Neural Networks were significantly better than Naive Bayes;
- Important performance indicators like precision, recall, and $f1$ were relevant to understand prediction performance. However, other studies did not use these indicators.
- Extensive testing is recommended;

(E) Besides this introduction, this paper is organized as follows. Section 2 presents the general background for delay prediction and concept drift. Section 3 presents the related work. Section 4 discusses the methodology (Stealthy Models) used for drift analysis over *SB* and *AB*. Section 5 presents experimental evaluation. Finally, Sect. 6 presents some concluding remarks and points out future work.

2 Background

The background is divided into two parts. Section 2.1 presents flight delays prediction using machine learning. Section 2.2 presents concept drift, including drift detection and handling.

2.1 Flight delays prediction using machine learning

Flight delay is a measure of the actual departure or arrival time minus their respective expected time. For the classification task of predicting whether a delay will occur or not, a threshold is set up to establish this binary variable. Most studies use a threshold of 15 min. It indicates that any flight 15 min or more late is marked as delayed (Gui et al. 2020; Guleria et al. 2019; Kim et al. 2016; Sternberg et al. 2016).

Many studies have been conducted and reported good results in the classification task (Belcastro et al. 2016; Moreira et al. 2018). In this context, random forests (*RF*) and neural networks (*NN*) achieved better *accuracy*. An *NN* is a bio-inspired computational approach that performs the processing of information through neurons that are connected through synapses (Han et al. 2011). Specifically, multi-layer perceptrons associated with long short term memory recurrent neural network (LSTM-RNN) and deep multilayer perceptrons seem to show good results during prediction (Gui et al. 2020; Kim et al. 2016; Rebollo and Balakrishnan 2014). Due to its interpretability, Naive Bayes (*NB*) is commonly included in the studies, as it encompasses a Baseline method. *NB* is a statistical classifier that can predict the probability of a tuple belonging to a particular class.

In a traditional classification problem, a dataset is separated into training and test sets. The model is built using the training set. It is common to partition the data set using a cross-validation strategy to optimize hyperparameters. Once the model is built, it is later evaluated using a test set (Han et al. 2011). This traditional approach is depicted in Fig. 1a.

(M) However, for flight delay prediction, the time dimension is relevant. Flights occur continuously each day as a streaming data source. Even when the entire streaming is stored in a single dataset, the time stamp of the flight events needs to be considered. It means that training should occur using past data to predict more recent data. It is depicted in Fig. 1b, where the i-th batch (training data) is used to build a model for further evaluation with more recent data at the next batch ($i + 1$) (test data). It is worth mentioning that the i-th batch corresponds to a sample of the dataset in the time interval

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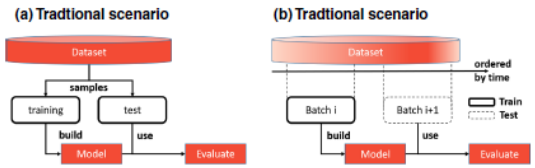
Referencial teórico

- Conceitos prévios necessários para o leitor entender o trabalho
  - De forma sucinta e suficiente
  - Em ordem de dependência
- Fontes
  - Livros de reputação
  - Artigos que sejam survey
    - Veja os artigos citados pelo survey
  - Artigos que tenha conceitos importantes

Fig. 1 Characterization of training and testing during classification: **a** traditional scenario; **b** streaming scenario



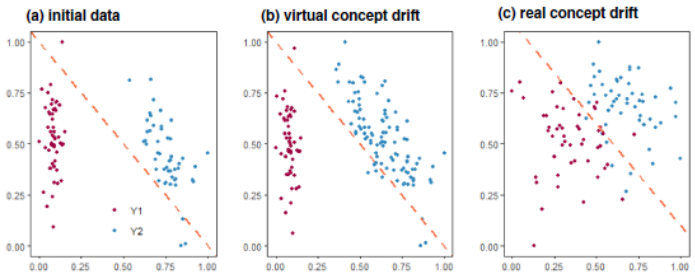


Fig. 2 Examples of types of drifts: **a** initial dataset; **b** virtual concept drift from **a**; **c** real concept drift from **a**

associated with i. While studying concept drift, such methodology is mandatory (Iwashita and Papa 2019).

Finally, metrics are used to evaluate the prediction performance of built models. Formally, two classes (positive and negative) were given: positive tuples corresponding to delays and negative tuples for the ones without it. P is the number of positive tuples, and N is the number of negative tuples. The class of the test set is compared to the class predicted by the built model, getting: True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN) (Han et al. 2011; Moreira et al. 2018). From these measures, it is possible to compute the most widely used metrics: $accuracy$ ($\frac{TP+TN}{P+N}$), $precision$ ($\frac{TP}{TP+FP}$), $recall$ ($\frac{TP}{TP+FN}$), and f_1 ($\frac{2 \times precision \times recall}{precision+recall}$).

2.2 Concept drift

Consider a classification problem, such that a set of input variables X is used to predict a class label Y. One of the main challenges when creating machine learning models is handling concept drift. It refers to a significant change in data distribution that interferes with the relation between the output class Y and input variables X. Formally, a concept at a time i is defined as the probability of the joint distribution χ of X and Y. It is described in Eq. (1). A concept drift between time i and j is defined as a difference (with statistical significance) between the probabilities $p_i(\chi)$ and $p_j(\chi)$. It is described in Eq. (2) (Webb et al. 2016).

$$concept_i = p_i(\chi) = p_i(X, Y) \quad (1)$$

$$p_i(\chi) \neq p_j(\chi) \quad (2)$$

The drifts can also be classified as real or virtual drifts (Iwashita and Papa 2019). Specifically, real concept drifts are defined by changes in the posterior probabilities $p(Y|X)$, commonly related to the class boundaries. Conversely, virtual concept drifts happen whenever the conditional probability $p(X)$ changes (Iwashita and Papa 2019; Lu et al. 2019; Hoens et al. 2012). A visual example is shown in Fig. 2.

Regarding flight delays, there can be relevant changes in the proportion of delayed flights. Some periods are more critical than others (Gui et al. 2020; Sternberg et al. 2016).

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Trabalhos relacionados

- Faça uma busca sistemática
  - Monte uma classificação
- Organize um texto comparativo baseado na classificação
  - Cada conceito da classificação merece, pelo menos, um parágrafo nos trabalhos relacionados citando os diversos trabalhos na categoria
- Os trabalhos diretamente associados podem necessitar de parágrafos próprios
- Elabore uma síntese que aponte para lacunas
  - Em especial aquelas abordados pelo método
- Faça um balanço entre o tamanho do artigo e o tamanho da seção de trabalhos relacionados

one, resembling the baseline strategy. The difference is that the decision is because no drift was observed. Conversely, continuous retraining may occur between each batch, resembling the *passive* strategy. Again, such a decision is based on whether drift is observed whenever a new batch is introduced (Iwashita and Papa 2019; Gama et al. 2014). Moreover, some techniques may change the size of the batches using soft computing (fuzzy systems) to identify drifts and adapt accordingly (Angelov and Zhou 2008; Lughofer and Angelov 2011; Zhou and Angelov 2007).

3 Related work

Past literature on delay prediction is extensive (Carvalho et al. 2021; Alla et al. 2021). We have carried out a systematic mapping study to identify relevant data science studies regarding flight delay prediction.[1] There were deemed only papers (Journals and Conferences) entirely written in English. The query was submitted to the Scopus database (considering titles, keywords, and abstracts) in December 2020 and returned 141 papers. Additionally, two studies were added through snowballing.

Regarding training and testing, the majority of studies uses traditional scenario (Fig. 1a) (Chen et al. 2008; Khanmohammadi et al. 2014; Alonso and Loureiro 2015; Yi et al. 2021; Alla et al. 2021). Specifically, besides the traditional 1-model scenario (Chen et al. 2008; Alonso and Loureiro 2015; Alla et al. 2021), model stacking was also investigated (Yi et al. 2021). However, this approach does not consider the possible drifts that are usually present on flight data. Studies that considered streaming approaches were deeply investigated, particularly those that contained drift handling strategies (Kim et al. 2016; Khamassi and Sayed-Mouchaweh 2014; Pesaranghader et al. 2016; Munoz Hernandez et al. 2019; Wang et al. 2019; Ai et al. 2019; Gui et al. 2020). In fact, only the work of Kim et al. (2016), Khamassi and Sayed-Mouchaweh (2014) and Pesaranghader et al. (2016) investigated the binary classifier problem.

Kim et al. (2016) used Deep Neural Networks to predict flight delays in the US flights using a *BSS* of 7 and 9 days before prediction. An additional Deep Recurrent Neural Network was trained to classify the critical delay status of each day. Since the scale used was of flight routes, these predictions were added to individual flight routes data and used as input in a Deep Neural Network. Although the *accuracy* was above 80% in most experiments, key metrics such as f_1, *precision*, and *recall* were not reported. As no specific strategy was used to identify or deal with drifts actively, we considered this approach a passive strategy.

Khamassi and Sayed-Mouchaweh (2014) proposed a new error active distance-based approach for drift detection and monitoring, named EDIST. Specifically, EDIST compares new data with existing data and retrains the model if significant changes are found in the distribution. The classifier used in this case was the Hoeffding Trees, and *BSS* was statistically adaptive. Active strategies of the drift detection method (DDM) and the early DDM (EDDM) are both error-based methods. They were implemented for comparison reasons. A baseline strategy of training with the first batch and predicting the remaining data was tested as well. They have applied these techniques to many synthetic and real-world datasets (including the US flight dataset). However, only the accuracy was used to report prediction performance.

Finally, Pesaranghader et al. (2016) tested many drift detection techniques with Naive Bayes and Hoeffding Trees. Specifically, their work proposes using the Hoeffding Inequality Theorem to test the difference in the probability of a given class between two *BSS* of 25 cases, characterizing an active strategy. DDM, EDDM, Adaptive Sliding Window (ADWIN), Hoeffding Drift Detection Method (HDDM), and Fast HDDM (FHDDM) active strategies were implemented for comparison reasons. ADWIN implements statistically adaptive batch sizes, defined whenever a drift is detected by average comparison. HDDM uses the Hoeffding inequality to compare distributions of batches. Finally, the proposed FHDDM uses Hoeffding inequality to compare errors from batches and thus detect drifts. The reported *accuracy* was around 65% for all experiments with aviation data. Fast Hoeffding Trees with Adaptive Windowing showed the best results for the US flight datasets.

To the best of our knowledge and considering the systematic mapping presented here, no study compared active and passive drift handling strategies for delay prediction in aviation. Moreover, the influence of the scale of the data used to train the models was never investigated as an essential factor for delay prediction. Besides accuracy, no other prediction performance indicators were reported as well.

4 Methodology

This paper aims to study drift handling strategies. Specifically, three different *active* strategies and one *passive* strategy were investigated. The selected techniques showed promising results in past studies but were never evaluated with flight data sets. Therefore, our analysis was based on this comparison (research question one). A second goal is to investigate drift handling strategies under the influence of different scales of training data: System-Based (*SB*) and Airport-Based (*AB*). In other words, how the scales interfere

[1] Search string used: ("flight delay") AND ("classification" OR "regression" OR "prediction").

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Metodologia

- Faz uso dos conceitos previamente necessários
- Apresenta de forma sucinta e suficiente a solução para o problema
  - arquitetura, organização de dados, processos, definições, algoritmos, prova de correção, análise de complexidade
- Apresenta as suas ideias originais
- Faz uso de figuras, diagramas e algoritmos
  - Abstrações da implementação
  - Cada objeto tem propósito claro e os seus itens presentes são todos apresentados

Table 1 Parameters used

Name	Description [values]
D	Dataset with flight and weather data
airport	Airport code for *AB* analysis; or *nil*, for *SB* analysis
mlm	Machine learning models
t	Time (yearly based time slices)
b	*BSS* for training
dd	Drift detection method
dh	Drift handling strategy

with drift handling strategies (research question two). It is worth mentioning that, to the best of our knowledge, the scale of training data was never analyzed in previous studies.

In this section, the general methodology (named Stealthy Models) used in the present work is described (Sect. 4). It comprises three steps: (1) preprocessing; (2) drift detection and handling; (3) model training and evaluation. Step 1 (Sect. 4.2) explains the preprocessing methods implemented. Step 2 (Sect. 4.3) consists of all drift-related strategies and techniques implemented. Finally, step 3 (Sect. 4.4) shows how the models are trained.The following subsections explain the general principle of Stealth Models and each step.

4.1 Stealthy models

Algorithm 1 describes the methodology used for flight delay prediction with concept drift. It requires seven parameters: *D*, *airport*, *mlm*, *t*, *b*, *dd*, and *dh*. The parameter *D* corresponds to the input dataset. The parameter *airport* identifies a single airport if the *AB* scale is used, or *nil* if the *SB* is used. Parameter *mlm* corresponds to the machine learning method. Parameter *t* is related to the time, and *b* corresponds to the size of the batch sequence. Finally, *dd* and *dh* correspond to the drift detection method and drift handling strategy, respectively. These parameters are described in Table 1.

Algorithm 1 General methodology

```
1: function Stealthy(D, airport, mlm, t, b, dd, dh)
2:     D_p ← preprocess(D)
3:     results ← ∅
4:     for each i ∈ t do
5:         train, D_i ← checkDrift(D_i, airport, t, b, dd, dh)
6:         mdl ← cloakModel(train, mlm, D_f)
7:         results ← results ∪ predictEvalDelays(mdl, D, t)
8:     return results
```

The first step of Stealthy 1 does the preprocessing (function *preprocess*). If *airport* is different from *nil*, the dataset *D* is filtered for that airport. Otherwise, it studies the entire *SB*. All steps of data preprocessing are described in Sect. 4.2. Line 4 implements a loop through all batches available within the dataset time frame available.

Lines 5 and 6 computes the drift detection method (*dd*), drift handling strategy (*dh*), and the two batch sequences (D_i and D_f). The mechanics of drift action are described in Sect. 4.3. The output of drift action is assigned to *train*, indicating if training is required at time *t*. If required, a machine learning method is trained using the batch sequence D_i (Line 6). The training process is described in Sect. 4.4. The output is a trained model (*mdl*). The model is stored for the selected *airport*, *mlm*, *dd*, and *dh*. If no training is required, previously trained model *mdl* is retrieved for the selected *airport*, *mlm*, *dd*, and *dh*. Line 7 adds the results from *predictEvalDelays* in a dataset. After batches are processed, the dataset with the results is returned.

4.2 Preprocessing

In our proposed workflow, the first step is to preprocess the data. The function *preprocess* expects a dataset with a binary delay feature and a DateTime feature to calculate the weekly proportion of delays used for drift checking. Then, the week number within the year is created to group and aggregate by the proportion of delayed flights. Statistical tests for drift detection are performed using the weekly values of delay proportion.

Regarding data cleaning, the *preprocessing* function removes cases with negative or too long durations (more than 19 h) as they are considered errors, considering this flight duration in BFS is not possible. Cases with missing delay values or estimated departure dates are also removed. For other features, mode imputation is used with categorical ones and mean for numeric data.

Finally, the preprocessing function returns a full preprocessed dataset with a binary delay feature. The new feature of the week number is also returned with the dataset. If an airport is passed, the dataset is reduced to the specific flights that departed from that airport. If *nil* is passed as airport, the entire dataset with all airports is returned.

4.3 Drift detection and handling

Regarding drift detection and handling, three methods and three strategies for drift detection and handling, respectively, were implemented. The drift handling strategies were the *baseline*, *active*, and *passive* drift handling strategies (Fig. 3). The *baseline* corresponds to training using the first batch sequence to predict all other batches. The *passive* strategy trained a model for each batch sequence to predict the next batch. Finally, the *active* strategy compared the current training batch sequence with the previous one to detect a drift.

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A metodologia deve responder as perguntas

- O modelo proposto viabiliza o desenvolvimento?
- O modelo proposto está bem escrito?
- O referencial teórico aborda todos os elementos do modelo?
- Existe elementos no modelo não mencionados no referencial teórico?
- O desenvolvimento da solução está de acordo com a modelagem?
- O desenvolvimento aborda o que será avaliado?
- O desenvolvimento resolve o problema proposto?
- O desenvolvimento está bem escrito?

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Fronteira entre revisão bibliográfica e metodologia

- Para observar o encadeamento, deve-se atentar ao sequenciamento lógico
  - Revisão Bibliográfica
    - Recomendável colocar as premissas em que o trabalho se apoia antes da explicação do abordagem
    - Termos gerais já existentes usado no trabalho
  - Metodologia
    - Termos novos inventados por você
- Isso facilita ao leitor identificar a sua contribuição

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Resultados

- Descreve a fonte de dados utilizada
- Descreve o roteiro experimental e sua execução
- Descreve as abordagens usadas para comparação
- Apresenta as avaliações acompanhadas de bons gráficos e tabelas
- Apresenta boas discussões:
  - Comparando as abordagens estudadas
  - Apresentando a diferença entre o resultado teórico esperado e o encontrado na prática
  - Explicando os motivos dos bons e maus desempenhos da proposta
  - Escrevendo coisas mais interessantes do que apenas legendar os gráficos
- A avaliação deve medir o núcleo metodológico
- É importante realizar análises "em cima" do que foi obtido após o emprego da abordagem
- Apresente os resultados de outros trabalhos com os quais pretende-se comparações

$$f1 = (\beta_0) + (\beta_1)x_1 + (\beta_2)x_2 + \cdots (\beta_i)x_i \quad (3)$$

Finally, each coefficient value is tested to check if it could be significant or by chance (Groß 2012). Significant betas may indicate that the corresponding parameter has a significant impact over $f1$ compared to its reference option.

5.4.2 Comparison of BSS

The Betas results of linear regression models are shown in Fig. 6. Regarding BSS, the linear regression analysis showed that only SBSV, SBSP, SBRJ, and SBBR had significant betas for BSS greater than one. In those cases, all the values were negative, meaning that using a BSS higher than one would mean a lower $f1$ score. Specifically, the decreases caused by these BSS options would be around one percent in $f1$. These results may indicate that increasing the size may not be relevant to the $f1$ score, and the 1-year size may be the best choice considering the values we tested. These results are shown in Fig. 6.

5.4.3 Comparison of classifiers

Using the same Regression analysis from Sect. 5.4 to understand the influence of each classifier in $f1$, the NB classifier was used as a reference. Among classifiers, KNN, RF, and NN were always significant when compared to the NB reference. Specifically, the KNN showed similar results to reference NB, with small positive and negative Betas. NN and RF were almost always better for $f1$ than NB and KNN.

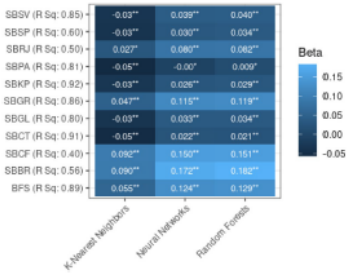
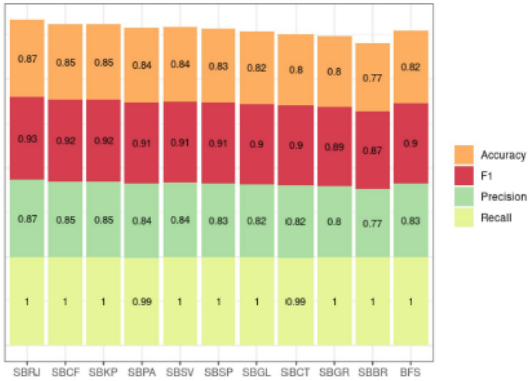


Fig. 7 Betas from linear regressions modeled for each classifier parameter used. $*p < 0.05$. $**P < 0.01$

The Beta for NN was negative only in SBPA airport, but all other cases showed significant positive values. Considering an RF/NN comparison, the difference in Betas was minimal, indicating that both may impact $f1$ in very similar ways.

These results may indicate that the classifier used may be necessary when predicting flight delays in BFS. Specifically, more complex models (NN and RF) may have better results. The differences between NN and RF classifiers were minor, and it may not be possible, with the present results, to ascertain that one of them has better results.

Fig. 8 Prediction performance metrics for airports



[1] R. Wazlawick, 2017, *Metodologia de Pesquisa para Ciência da Computação*. Elsevier Brasil.
[2] J. Zobel, 2015, *Writing for Computer Science*. Springer.
[3] L. Giusti, L. Carvalho, A.T. Gomes, R. Coutinho, J. Soares, and E. Ogasawara, 2022, Analyzing flight delay prediction under concept drift, *Evolving Systems*

Conclusões

- Resumem as contribuições do trabalho
- Resumem os principais resultados alcançados e suas implicações
- Indicam o escopo/limitações
- Apontam para trabalhos futuros

strategies may train more than one time, the overall increase in execution time was not significant compared to baseline for most airports (Fig. 9).

Considering Execution Time, the regression analysis show that the training time tends to be more significant as the BSS increases. This result is expected considering that the number of data points increases. However, considering that BSS higher than one does not significantly increase the prediction performance, an essential gain in execution time is possible.

Regarding execution performance, all models increased the execution time significantly when compared to NB. Specifically, NN was the slower option. Moreover, the Random Forests model shows betas of a quarter of NN's. These results may indicate that RF can be faster than NN, which is an essential game-changer since these two classifiers showed similar results in prediction performance (Fig. 10).

6 Conclusions

In this paper, we analyzed different types of drift handling strategies in aviation. Two research questions were answered to achieve the main objective of this study: (1) how do drift handling strategies influence the prediction performance of delays? (2) Do different scales change the results of drift handling strategies? We analyzed different strategies and scales and compared performances with linear regressions and a top-k analysis (Fig. 10).

It was observed that drift handling strategies are relevant. Their impact varies according to the scales used. The experimental evaluation was done using a dataset that integrates weather and flight data from the Brazilian system. Moreover, the *passive* and *active* strategies showed better recall scores. For $f1$ scores, the strategies had similar results, with the Passive strategy showing slightly better results. It may be related to the high prevalence of drifts. In this case, strategies that always retrain machine learning models offer better results than those that train only once. However, extensive testing is recommended. Nonetheless, choosing machine learning models may have a higher impact on $f1$ than drift handling strategies.

As limitations, the first one is the drift detection methods used in the experimental evaluation. They were focused only on changes $P(Y)$. Future studies may consider testing error-based drift handling strategies, investigating ensemble drift detection methods, implementing more robust classifiers (CNN, LSTM, and others), check drift handling strategies under different thresholds for flight delays, and evaluate other metrics such as the ROC curve.

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Author Contributions All authors contributed equally to the study. EO conceptualized the study design. LG and LC acquired the data, conducted data analysis and interpretation. AG, RC, JS revised it critically for intellectual content. All authors have the approval of the final version.

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Declarations

Conflict of interests The authors declare that they have no competing interests.

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Refinamento

Processo de escrita

Composição do texto

- Desenvolva o planejamento do texto
  - Monte uma apresentação de estruturação de texto
  - Não é apresentação final do trabalho
- Comece trabalhando em uma das seções
  - Métodos, Resultados, Trabalhos Relacionados, Introdução (DP)?
  - Oriente a ordem a partir da mitigação de riscos
- Preencha um rascunho
  - Explique os conceitos
  - Adicione a revisão bibliográfica
  - Organize o fluxo lógico
- Refina o texto
  - Corrija os erros
  - Atente ao layout

[1] J. Zobel, 2015, *Writing for Computer Science*. Springer.
[2] R. Wazlawick, 2017, *Metodologia de Pesquisa para Ciência da Computação*. Elsevier Brasil.

Organize os parágrafos

- Cada parágrafo descreve apenas uma ideia central (tópico)
  - Havendo mais de um tópico, quebre o parágrafo em partes
- Cada parágrafo tem, pelo menos, três frases
- Cada seção tem, pelo menos, três parágrafos
- Cada capítulo pode ou não ter seção, ou, pelo menos, três seções

[1] J. Zobel, 2015, *Writing for Computer Science*. Springer.
[2] R. Wazlawick, 2017, *Metodologia de Pesquisa para Ciência da Computação*. Elsevier Brasil.

Encadeamento

- A ideia central de um parágrafo deve levar suavemente a ideia central do próximo parágrafo
- Quando há um rompante de ideias:
  - Ordem dos parágrafos está errada
  - Falta um ou mais parágrafos para realizar a transição

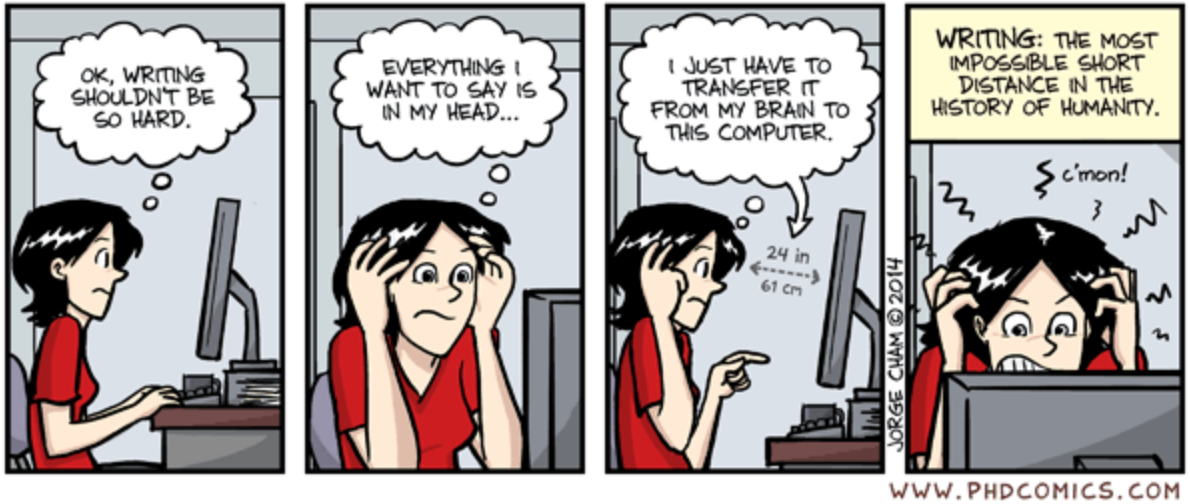
[1] J. Zobel, 2015, *Writing for Computer Science*. Springer.
[2] R. Wazlawick, 2017, *Metodologia de Pesquisa para Ciência da Computação*. Elsevier Brasil.

Preparação para começar a escrever

- Monte uma apresentação do seu trabalho
  - Modelar o que deve ser escrito
- Crie slides para cada elemento indicado no slide “Estrutura geral de um texto científico”
- Cada "**item**" no slide é uma ideia central do texto que você vai montar
  - Ele vai significar um parágrafo do seu texto
  - A ordem dos "itens" define o encadeamento das ideias
  - Apresente para você mesmo e reflita sobre o encadeamento da sua apresentação
  - Apresente para o seu orientador para ele te ajudar a reorganizar o encadeamento e foco

[1] J. Zobel, 2015, *Writing for Computer Science*. Springer.
[2] R. Wazlawick, 2017, *Metodologia de Pesquisa para Ciência da Computação*. Elsevier Brasil.

Escreva


OK, WRITING SHOULDN'T BE SO HARD.
EVERYTHING I WANT TO SAY IS IN MY HEAD...
I JUST HAVE TO TRANSFER IT FROM MY BRAIN TO THIS COMPUTER.
24 in
61 cm
WRITING: THE MOST IMPOSSIBLE SHORT DISTANCE IN THE HISTORY OF HUMANITY.
c'mon!
JORGE CHAM © 2014
WWW.PHDCOMICS.COM

Revise de modo crítico

- Uma vez que a versão inicial do texto foi concebida, procure revisar o texto um ou dois dias depois
  - O afastamento do texto ajuda a condução de uma revisão crítica
- Quanto mais cuidadosa e rigorosa for a correção de seu texto, mais você se esforçará para produzir um trabalho cada vez mais claro, objetivo e com argumentos convincentes

[1] J. Zobel, 2015, *Writing for Computer Science*. Springer.
[2] R. Wazlawick, 2017, *Metodologia de Pesquisa para Ciência da Computação*. Elsevier Brasil.

Roteiro para revisar o seu artigo

- Confira o resumo
- Confira a introdução
- Confira as conclusões
- Confira a estrutura
  - Cada seção tem, pelo menos, três parágrafos
  - Cada parágrafo tem, pelo menos, três frases
  - Cada parágrafo tem um ideia central
  - Analise o encadeamento do texto
- Confira a qualidade das referências
- Confira a proporção de páginas

[1] J. Zobel, 2015, *Writing for Computer Science*. Springer.
[2] R. Wazlawick, 2017, *Metodologia de Pesquisa para Ciência da Computação*. Elsevier Brasil.

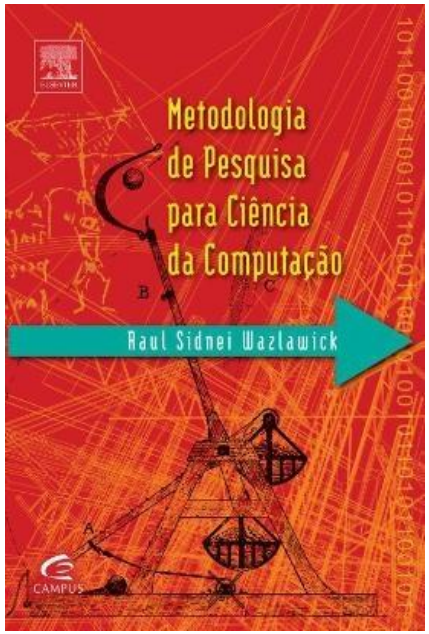
Conferência da consistência entre as seções

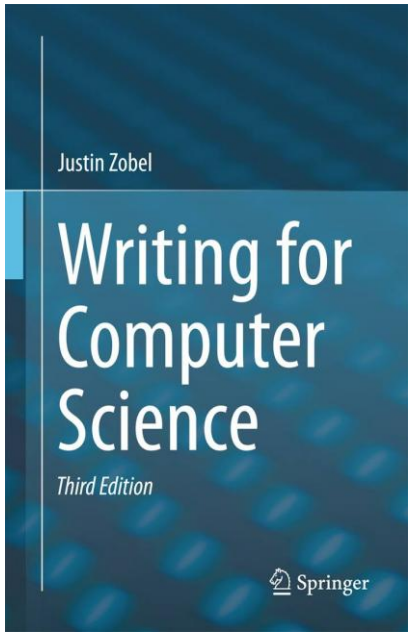
- Analise a contribuição de todas as seções na formula dos pontos fortes e fracos do trabalho
- Verifique se a introdução faz uma boa síntese do trabalho
  - Se tem a motivação, definição do problema, apresentação da solução, contribuição
- Verifique se o resumo faz uma bom síntese da introdução
- Verifique se o referencial teórico apoia a metodologia
- Verifique se a metodologia apoia o experimento
- Verifique se a metodologia e experimentos respondem às perguntas de pesquisa

Faça estas verificações usando o chatgpt

[1] J. Zobel, 2015, *Writing for Computer Science*. Springer.
[2] R. Wazlawick, 2017, *Metodologia de Pesquisa para Ciência da Computação*. Elsevier Brasil.

Referências





Leitura adicional: Metodologia de Pesquisa para Ciência da Computação, Capítulos 8, 9 e 11 e Writing for Computer Science – Capítulo 5, 6, 7, 8 e 13

Slides e vídeos em: https://eic.cefet-rj.br/~eogasawara/metodologia-cientifica/

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[2] R. Wazlawick, 2017, *Metodologia de Pesquisa para Ciência da Computação*. Elsevier Brasil.